

Lecture 29-VLAN Trunking, Native VLAN & Router on a Stick (ROAS)

1 Access Port vs Trunk Port

When configuring VLAN networks, **switch ports must be set to the correct mode.**

Access Port

An **Access Port** carries traffic for **only ONE VLAN**.

✓ Used for **end devices**

✓ Example devices:

- PCs 
- Printers 
- IP Phones 
- Cameras 

Example configuration:

```
switchport mode access  
switchport access vlan 10
```

Trunk Port

A **Trunk Port** carries traffic for **MULTIPLE VLANs** over a single interface.

✓ Used between:

- Switch ↔ Switch
- Switch ↔ Router
- Switch ↔ Layer 3 Switch

Example configuration:

```
switchport mode trunk
```

Important Rule

Device Type	Port Mode
End Device (PC)	Access Mode
Network Device (Switch/Router)	Trunk Mode

Very important CCNA concept

Example VLAN Topology

Imagine a network with **two switches**.

PCs → SW1 → SW2 → Router

Engineering department exists in **two locations**:

- Some PCs connected to **SW1**
- Some PCs connected to **SW2**

All belong to:

VLAN 10

Therefore the **switch-to-switch link must carry VLAN 10 traffic**.

This is why **TRUNK ports are required**.

Why Trunk Ports Are Needed

In small networks we could connect switches like this:

VLAN10 → Interface 1

VLAN20 → Interface 2

VLAN30 → Interface 3

But in real networks:

- ✗ Too many VLANs
- ✗ Wastes switch interfaces
- ✗ Routers may not have enough ports

Instead we use **TRUNK PORTS**.

Trunk Port Advantage

One interface can carry **multiple VLAN traffic simultaneously**.

Example:

Gig0/1 carries VLAN 10, 20, 30

2 VLAN Tagging

Question:

 If many VLANs share one trunk link, how does the switch know which VLAN a frame belongs to?

Answer:

VLAN TAG

Switch **adds a tag to the Ethernet frame**.

This tag contains the **VLAN ID**.

Tagged vs Untagged Frames

Port Type	Frame Type
Trunk Port	Tagged
Access Port	Untagged

 Important concept for CCNA exam.

**3**

VLAN Trunking Protocols

There are **two trunk protocols**.

Protocol	Description
ISL	Cisco proprietary
IEEE 802.1Q	Industry standard

● ISL (Inter-Switch Link)

- Cisco proprietary
- Older protocol
- Rarely used today

● IEEE 802.1Q (Dot1Q)

- Industry standard
- Supported by almost all vendors
- Used in real networks

📌 **For CCNA you mainly study 802.1Q**

**4**

Ethernet Frame with 802.1Q Tag

802.1Q inserts a **4-byte VLAN tag** into the Ethernet frame.

The tag is inserted between:

Source MAC

and

Type/Length

Tag size:

4 bytes (32 bits)

✖ Components of 802.1Q Tag

The tag has two main sections.

1 TPID – Tag Protocol Identifier

- Size: **16 bits**
- Value: **0x8100**

This tells the switch:

This frame uses 802.1Q tagging

2 TCI – Tag Control Information

TCI contains **three fields**.

◆ PCP (Priority Code Point)

Size:

3 bits

Used for:

Class of Service (CoS)

Allows prioritizing important traffic.

Example:

- Voice calls 📞
 - Video streaming 🎥
-

◆ DEI (Drop Eligible Indicator)

Size:

1 bit

Used when network is congested.

Frames marked with DEI may be dropped.

◆ VID (VLAN ID)

Size:

12 bits

This field identifies the VLAN.

VLAN ID Range

12 bits allow:

$2^{12} = 4096$ VLANs

Range:

0 – 4095

However:

VLAN ID Status

0 Reserved

4095 Reserved

Usable VLANs:

1 – 4094

 Important CCNA fact.

5 Inter-VLAN Routing

Devices in different VLANs **cannot communicate directly**.

Example:

Sales VLAN (30)

HR VLAN (20)

Switch **cannot route traffic between them**.

Traffic must go through:

Router

This process is called:

INTER-VLAN ROUTING



Packet Flow Example

Example:

Sales PC → HR PC

Steps:

- 1 Packet goes to switch
- 2 Switch sees destination VLAN is different
- 3 Switch forwards packet to router
- 4 Router checks routing table
- 5 Router sends packet back to switch
- 6 Switch forwards to HR PC

📌 Important rule:

Switch never routes between VLANs

Only **router or Layer-3 switch** can do routing.



6 Native VLAN

802.1Q includes a feature called:

Native VLAN

Default native VLAN:

VLAN 1

Important behavior:

Frames in the **native VLAN are NOT tagged**.

Native VLAN Example

If a switch receives an **untagged frame on a trunk port**:

Switch assumes:

Frame belongs to native VLAN

Native VLAN Mismatch

If switches use **different native VLANs**, problems occur.

Example warning:

CDP-4-NATIVE_VLAN_MISMATCH

Security Best Practice

Network engineers recommend:

Change native VLAN to an unused VLAN

Example:

VLAN 1001

Reason:

Prevents VLAN hopping attacks.

7 Router on a Stick (ROAS)

Router on a Stick allows **multiple VLANs to use one router interface**.

Instead of multiple physical interfaces, we use:

SUBINTERFACES

Each subinterface represents one VLAN.

Example:

g0/0.10

g0/0.20

g0/0.30

Each has:

- VLAN tag
- IP address



How ROAS Works

Router receives tagged frames.

Router checks VLAN tag.

Router forwards traffic through the correct subinterface.



VLAN Trunk + ROAS LAB



Devices Used

- 2 Switches
 - 1 Router
 - 7 PCs
-



VLAN Design

VLAN Department Network

10	Engineering	10.0.0.0/26
20	HR	10.0.0.64/26
30	Sales	10.0.0.128/26



Step 1 – Create VLANs on Switches

SW1 Configuration

```
SW1> enable
```

```
SW1# configure terminal
```

```
vlan 10  
name Engineering
```

```
vlan 20  
name HR
```

```
vlan 30  
name Sales
```

```
vlan 1001  
name Native
```

Assign PC Ports (Access Mode)

```
interface fa0/1
switchport mode access
switchport access vlan 10
```

```
interface fa0/2
switchport mode access
switchport access vlan 10
```

```
interface fa0/3
switchport mode access
switchport access vlan 30
```

```
interface fa0/4
switchport mode access
switchport access vlan 30
```

Verify:

```
show vlan brief
```

SW2 Configuration

```
interface fa0/1
switchport mode access
switchport access vlan 20
```

```
interface fa0/2
switchport mode access
switchport access vlan 10
```

```
interface fa0/3
switchport mode access
switchport access vlan 10
```



Step 2 – Configure Trunk Between Switches

SW1

```
interface g0/1
switchport mode trunk
switchport trunk allowed vlan 10,20,30
switchport trunk native vlan 1001
```

Verify:

```
show interface trunk
```

SW2

```
interface g0/1
switchport mode trunk
switchport trunk allowed vlan 10,20,30
switchport trunk native vlan 1001
```

Trunk to Router

```
interface g0/2
switchport mode trunk
switchport trunk allowed vlan 10,20,30
switchport trunk native vlan 1001
```



Step 3 – Router on a Stick Configuration

Enable Router Interface

```
R1> enable
R1# configure terminal
```

```
interface g0/0
no shutdown
```

Subinterface for VLAN 10

```
interface g0/0.10
encapsulation dot1q 10
ip address 10.0.0.62 255.255.255.192
```

Subinterface for VLAN 20

```
interface g0/0.20
encapsulation dot1q 20
ip address 10.0.0.126 255.255.255.192
```

Subinterface for VLAN 30

```
interface g0/0.30
encapsulation dot1q 30
ip address 10.0.0.190 255.255.255.192
```

Verify Router Interfaces

```
show ip interface brief
```

Step 4 – Connectivity Test

Test ping between VLANs.

Example:

PC1 ping 10.0.0.65

If successful:

✓ Inter-VLAN routing works.

Packet Tracer Testing

Use:

Simulation Mode

Steps:

- 1 Enable ICMP filter
- 2 Send ping
- 3 Use **Capture/Forward**

You will see packets travel:

PC → Switch → Router → Switch → PC



Step-by-Step Lab Summary

Step 1

Create VLANs on switches.

Assign PC ports as **Access ports**.

Step 2

Configure **Trunk ports** between switches.

Allow only required VLANs.

Set **unused VLAN as native VLAN**.

Step 3

Configure **Router on a Stick**.

Create subinterfaces.

Assign gateway IP addresses.

Step 4

Test connectivity using **Ping**.

Real-World Example

Large company network:

Departments:

- Engineering
- HR
- Sales

Each department uses **separate VLANs**.

But only **one trunk link** connects switches.

Router performs **inter-VLAN routing**.

This reduces:

- ✓ Cable usage
 - ✓ Hardware cost
 - ✓ Network complexity
-

Interview One-Liners

Trunk Port

“A trunk port carries traffic for multiple VLANs using VLAN tagging.”

802.1Q

“802.1Q is an IEEE standard used for VLAN tagging on trunk links.”

Router on a Stick

“Router on a Stick uses router subinterfaces to route traffic between multiple VLANs using a single physical interface.”

Final Key Points

- ✓ Access ports carry **one VLAN**
- ✓ Trunk ports carry **multiple VLANs**
- ✓ 802.1Q adds **VLAN tags**
- ✓ VLAN ID range = **1 – 4094**
- ✓ Native VLAN default = **1**
- ✓ Untagged frames belong to **native VLAN**
- ✓ Inter-VLAN communication requires **router or Layer 3 switch**
- ✓ Router on a Stick uses **subinterfaces**

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Chapter Summary

◆ Access Port

An **Access Port** is a switch port that carries **traffic for only one VLAN**.



Used for **end devices** such as:

-  PCs
-  Printers
-  IP Phones
-  Cameras

Example configuration:

```
switchport mode access  
switchport access vlan 10
```

Important points:

- ✓ One VLAN only
- ✓ Frames are **untagged**
- ✓ Used for **end devices**



Trunk Port

A **Trunk Port** is a switch port that carries **traffic for multiple VLANs simultaneously**.



Used between:

- Switch ↔ Switch
- Switch ↔ Router
- Switch ↔ Layer-3 Switch

Example configuration:

```
switchport mode trunk
```

Example:

Interface VLANs

Gig0/1 10, 20, 30

This means the interface carries **traffic for multiple VLANs**.

VLAN Tagging

When multiple VLANs share one trunk link, the switch must identify which VLAN the frame belongs to.

This is done using:

VLAN Tagging

The switch inserts a **VLAN tag inside the Ethernet frame**.

This tag includes the **VLAN ID**.

Tagged vs Untagged Frames

Port Type	Frame Type
Access Port	Untagged
Trunk Port	Tagged

 Exception: **Native VLAN frames are untagged even on trunk ports**

VLAN Trunking Protocols

There are two trunking protocols.

Protocol Description

ISL Cisco proprietary

802.1Q Industry standard

● ISL (Inter-Switch Link)

- Cisco proprietary protocol
 - Old technology
 - Rarely used today
-

● IEEE 802.1Q (Dot1Q)

- Industry standard
 - Used in almost all modern networks
 - Inserts **4-byte VLAN tag**
-

802.1Q Tag Structure

802.1Q adds **4 bytes** to Ethernet frame.

Main components:

Field Size	Purpose
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TPID 16 bits	Tag protocol identifier
--------------	-------------------------

PCP 3 bits	Priority
------------	----------

DEI 1 bit	Drop eligibility
-----------	------------------

VID 12 bits	VLAN ID
-------------	---------

VLAN ID Range

VLAN ID Status

0	Reserved
---	----------

1	Default VLAN
---	--------------

4095	Reserved
------	----------

Usable VLANs:

Inter-VLAN Routing

Devices in different VLANs **cannot communicate directly**.

Example:

- VLAN 10 (Engineering)
- VLAN 20 (HR)

A **Layer-2 switch cannot route between them**.

Traffic must go through:

- ✓ Router
- ✓ Layer-3 switch

This process is called:

Inter-VLAN Routing

Native VLAN

The **Native VLAN** is a special VLAN on trunk links.

Important behavior:

- ✓ Frames from native VLAN are **NOT tagged**

Default native VLAN:

VLAN 1

Native VLAN Mismatch

If two switches use **different native VLANs**, this error occurs:

CDP-4-NATIVE_VLAN_MISMATCH

This can cause:

- ✗ VLAN leakage
 - ✗ Security problems
-

Security Best Practice

Change native VLAN to **unused VLAN**.

Example:

VLAN 1001

Purpose:

- ✓ Prevent VLAN hopping attacks
-

Router on a Stick (ROAS)

Router on a Stick allows **multiple VLANs to use one router interface**.

Instead of using multiple physical ports, the router creates **subinterfaces**.

Example:

Subinterface VLAN

g0/0.10 VLAN 10

g0/0.20 VLAN 20

g0/0.30 VLAN 30

Each subinterface has:

- ✓ VLAN tag
 - ✓ Gateway IP address
-

How ROAS Works

- 1 PC sends packet
- 2 Switch forwards packet to router trunk
- 3 Router receives tagged frame

- 4 Router routes packet between VLANs
 - 5 Packet returned to switch
 - 6 Switch forwards to destination
-

Chapter Conclusion

This chapter explains **how VLAN traffic travels across switches and routers**.

Key ideas:

- ✓ Access ports connect end devices
 - ✓ Trunk ports carry multiple VLANs
 - ✓ 802.1Q tagging identifies VLANs
 - ✓ Native VLAN frames are untagged
 - ✓ Different VLANs require routing
 - ✓ Router on a Stick enables inter-VLAN routing using one interface
-  In real enterprise networks, **VLAN trunking and ROAS are fundamental concepts for scalable network design.**
-

Q & A

1 What is a Trunk Port?

Answer

A **Trunk Port** is a switch port that carries **traffic for multiple VLANs simultaneously**.

It uses **VLAN tagging (802.1Q)** to identify which VLAN the frame belongs to.

Example use cases:

- Switch to Switch connection
- Switch to Router connection

Example configuration:

```
interface g0/1
switchport mode trunk
```

2 What is an Access Port?

Answer

An **Access Port** is a switch port that belongs to **only one VLAN**.

It connects **end devices**.

Examples:

- PC
- Printer
- IP phone

Configuration example:

```
switchport mode access
switchport access vlan 10
```

3 What is VLAN Tagging?

Answer

VLAN tagging is a process where the switch inserts a **VLAN identifier into an Ethernet frame**.

This tag allows trunk links to carry **multiple VLAN traffic simultaneously**.

The tagging protocol used is:

IEEE 802.1Q

4 What is IEEE 802.1Q?

Answer

IEEE 802.1Q is the industry standard protocol used for **VLAN tagging on trunk links**.

It inserts a **4-byte VLAN tag inside Ethernet frames**.

Purpose:

- ✓ Identify VLAN ID
 - ✓ Support multiple VLANs on one link
-

5 What is ISL?

Answer

ISL (Inter-Switch Link) is a Cisco proprietary VLAN tagging protocol.

Characteristics:

- Developed by Cisco
- Encapsulates Ethernet frames
- Rarely used today

Modern networks use **802.1Q instead**.

6 What is the Native VLAN?

Answer

The **Native VLAN** is a special VLAN on trunk ports where frames are sent **without VLAN tags**.

Default native VLAN:

VLAN 1

Important rule:

Untagged frames on trunk ports belong to **Native VLAN**.

7 What is Native VLAN Mismatch?

Answer

Native VLAN mismatch occurs when two trunk ports use **different native VLANs**.

Example:

Switch Native VLAN

SW1 1

SW2 100

This causes:

- ✗ Network errors
- ✗ VLAN leakage
- ✗ Security risks

Error message example:

CDP-4-NATIVE_VLAN_MISMATCH

8 What is Router on a Stick?

Answer

Router on a Stick (ROAS) is a method used for **inter-VLAN routing** using a **single router interface**.

Instead of using multiple physical interfaces, the router creates **subinterfaces**.

Example:

```
interface g0/0.10
encapsulation dot1q 10
ip address 10.0.0.62 255.255.255.192
```

Each subinterface represents one VLAN.

9 Why Router on a Stick is used?

Advantages:

- ✓ Saves router interfaces
- ✓ Reduces hardware cost
- ✓ Simplifies network design
- ✓ Supports multiple VLANs

10 What is Inter-VLAN Routing?

Answer

Inter-VLAN routing is the process of **allowing communication between different VLANs**.

Example:

Engineering VLAN → HR VLAN

Devices in different VLANs require:

- ✓ Router
 - ✓ Layer-3 Switch
-

1 1 What command verifies trunk configuration?

show interface trunk

Displays:

- Trunk interfaces
 - Allowed VLANs
 - Native VLAN
-

1 2 What command shows router subinterfaces?

show ip interface brief

This shows:

- ✓ Interface status
 - ✓ IP addresses
 - ✓ Subinterfaces
-

1 3 Troubleshooting Question

Question

PC in VLAN 10 cannot ping PC in VLAN 20. Why?

Possible reasons:

- 1** Router not configured
 - 2** Trunk not configured
 - 3** VLAN not allowed on trunk
 - 4** Incorrect gateway on PCs
 - 5** Router subinterface missing
-

Interview Scenario Question

Question

Why can't a switch route traffic between VLANs?

Answer

A **Layer-2 switch operates at OSI Layer 2** and forwards frames using **MAC addresses only**.

Routing requires:

- IP processing
- Layer 3 functionality

Therefore **router or Layer-3 switch is required**.

Important Tip

Explain Router on a Stick in one line

Answer:

“Router on a Stick is a method of inter-VLAN routing where a single router interface uses multiple subinterfaces with 802.1Q tagging to route traffic between VLANs.”

Final Key Points to Remember

- ✓ Access ports carry **single VLAN**
 - ✓ Trunk ports carry **multiple VLANs**
 - ✓ 802.1Q adds **4-byte VLAN tag**
 - ✓ VLAN ID range = **1–4094**
 - ✓ Native VLAN frames are **untagged**
 - ✓ Native VLAN default = **VLAN 1**
 - ✓ Inter-VLAN routing requires **router or L3 switch**
 - ✓ Router on a Stick uses **subinterfaces**
-